MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6C21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, pulse amplifier

See figure 1

Weight: 1.3 pounds (589.7 grams) nominal

Mounting position: Vertical

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	ib	ic	Pp	Pg
Unit:	V	kVdc	kVdc	a	a	W (Note 1)	W
Maximum:	8.5	30	-2	15	6	300	50
TEST CONDITIONS:	8.2 Vac	27	---	---	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	---		0.65	II	If	15.9	17.7	Aac
1231	Peak emission	---	eb = ec = 3,000 v (max)	0.65	II	is	20	---	a
1266	Primary grid emission	---	Ic = 240 mAdc; Ef = 9.0 Vac; t = 15 sec (max)	0.65	II	Isg	---	-500	μAdc
---	Pulse operation (1)	3	Eb = 3 kVdc; Ec = -400 Vdc; esig = 2,000 v (max)/ib = 15 a	0.65	II	ic	---	5	a
---	Pulse operation (2)	4	Ec = -1,500 Vdc; esig = 3,100 v (max)/ib = 15 a (min); R1 = 1,600; Ebb = 27 kVdc (min)	0.65	II	---	---	---	---
1256	Electrode current (anode)	---	Ec = -1,200 Vdc	0.65	II	Ib	---	0.5	mAdc
1266	Total grid current	2	Eb = 2.0 kVdc; Ec/Ib = 180 mAdc; t = 120	0.65	II	Ic	---	-20	μAdc
1261	Electrode voltage (grid)	---	Eb = 2.0 kVdc; Ec/Ib = 180 mAdc	0.65	II	Ec	-17.5	-27.5	Vdc
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	---		---	---	Cgp	3.0	5.6	pF
						Cin	7.0	12.0	pF
						Cout	---	2.0	pF
	Quality conformance inspection, part 3								
---	Life test	---	Group C; pulse operation (2); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
---	Pulse operation (1)	---		---	---	ic	---	5	a
1266	Primary grid emission	---		---	---	Isg	---	-500	μAdc
1231	Peak emission	---		---	---	is	15	---	a
1031	Low-frequency vibration	5	No voltages	---	---	---	---	---	---
1036	Bump	5	Angle = 10°	---	---	---	---	---	---
---	Low-frequency vibration and bump end points:								
1266	Total grid current	---		---	---	Ic	---	-20	μAdc
1261	Electrode voltage (grid)	---		---	---	Ec	-17.5	-27.5	Vdc

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TES:

1. The product of the pulse duration in seconds and the pulse recurrence frequency in pulses per second shall not exceed 0.002. During any 100-microsecond interval, the maximum operating time shall not exceed 5 microseconds.

(E) 2. This test is to be the first test performed at the conclusion of the holding period.

3. A load resistor (noninductive) of 5 ohms shall be used in conjunction with a synchroscope to measure the anode current. The specification of the pulse shall be the same as in note 4. The grid pulse current shall be measured by a suitable noninductive resistor (in series with the grid circuit) and a synchroscope. The grid signal pulse amplitude shall be measured by suitable means.

4. The tube shall be operated as a pulse amplifier with a noninductive resistor of 1,600 ohms or less as the net anode load. Tubes may be tested in any number, provided the anode and grid resistors are modified accordingly. The pulse produced across the resistor shall have the following characteristics:

   (a) Minimum pulse duration: 1.0 microsecond, measured at the top of the pulse.

   (b) Minimum recurrence frequency: 1,500 pulses per second.

   (c) Duty: Product of (a) and (b) shall be 0.002, or greater. During this test there shall be no evidence of internal sparking or flashover for a period of 2 minutes.

5. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

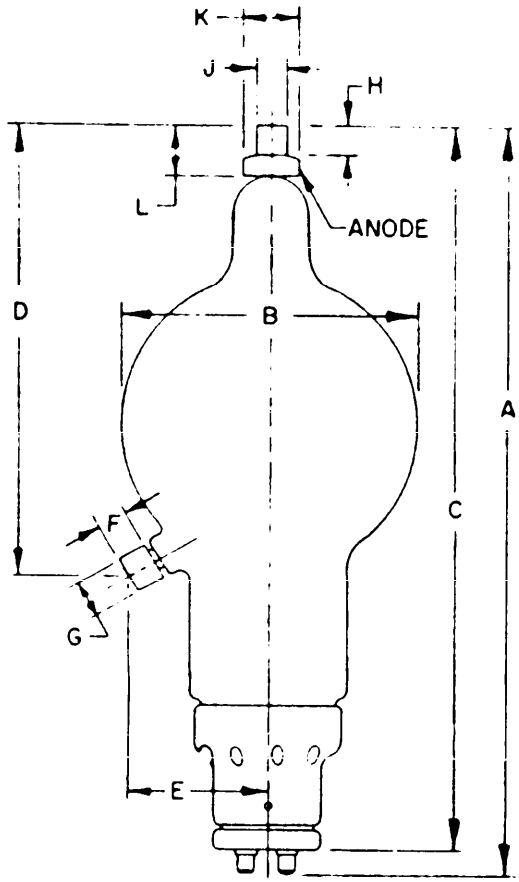
User activities:
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Air Force - 11

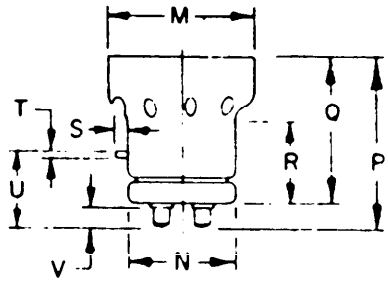
Preparing activity:
Navy - EC

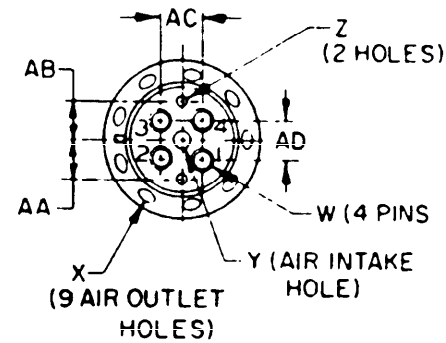
Agent:
DSA - ES

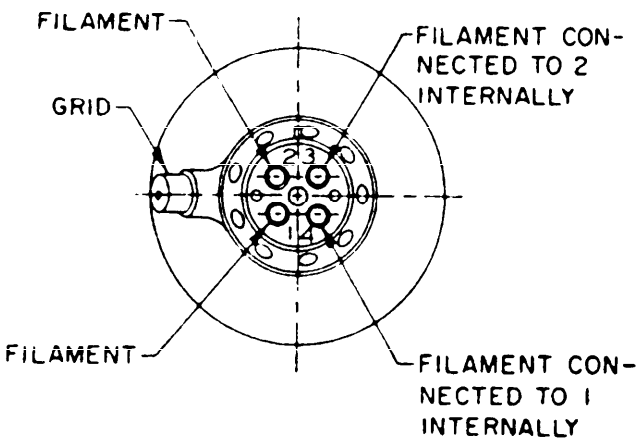
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	11.875(301.63)	12.625(320.68)	
B		5.125(130.18)	
D	7.000(177.80)	8.000(203.20)	
E	2.000 (50.80)	2.500 (63.50)	
G	.559 (14.20)	.573 (14.55)	
J	.559 (14.20)	.573 (14.55)	
Ⓔ Quality conformance inspection, part 3			b
C	11.563(293.70)	12.313(312.75)	
F	.484 (12.29)		
H	.484 (12.29)		
N	1.840 (46.74)	1.867 (47.42)	
P	2.656 (67.46)	2.719 (69.06)	a
Q	2.375 (60.33)	2.438 (61.93)	
R	1.313 (33.35)		
S	.075 (1.91)	.110 (2.79)	
T	.078 (1.98)	.082 (2.08)	
U	1.145 (29.08)	1.185 (30.10)	a
V	.250 (6.35)	.320 (8.13)	a
W	.184 (4.67)	.190 (4.83)	
Y	.380 (9.65)	.390 (9.91)	
AC	.680 (17.27)	.700 (17.78)	
AD	.680 (17.27)	.700 (17.78)	
Reference dimensions			
K	.875(22.23)		
L	.844(21.44)		
M	2.500(63.50)		
X	.250 (6.35)		
Z	.250 (6.35)		
AA	.594(15.09)		
AB	.594(15.09)		

FIGURE 1. Outline drawing of electron tube type 6C21.

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NOTES:
a. On finished tube, add .060 inch (1.52 mm) maximum for solder.
b. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6C21 - Continued.

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